

Firewall Rule – Python Implementation

```
from http.server import BaseHTTPRequestHandler, HTTPServer
```

```
BLOCKED_PATHS = ["/tomcatwar.jsp"]
```

```
BLOCKED_HEADERS = {
```

```
    "suffix": "%>/",
```

```
    "c1": "Runtime",
```

```
    "c2": "<",
```

```
    "DNT": "1",
```

```
    "Content-Type": "application/x-www-form-urlencoded"
```

```
}
```

```
BLOCKED_QUERY = "?pwd=j&cmd="
```

```
class FirewallHandler(BaseHTTPRequestHandler):
```

```
    def do_GET(self):
```

```
        if self.is_malicious():
```

```
            self.send_response(403)
```

```
            self.end_headers()
```

```
            self.wfile.write(b"Blocked by firewall rule")
```

```
        else:
```

```
            self.send_response(200)
```

```
            self.end_headers()
```

```
            self.wfile.write(b"Request allowed")
```

```
    def do_POST(self):
```

```
        if self.is_malicious():
```

```
            self.send_response(403)
```

```
            self.end_headers()
```

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```
        self.wfile.write(b"Blocked by firewall rule")
    else:
        self.send_response(200)
        self.end_headers()
        self.wfile.write(b"Request allowed")

    def is_malicious(self):
        # Check path
        if self.path in BLOCKED_PATHS:
            return True

        # Check query string
        if BLOCKED_QUERY in self.path:
            return True

        # Check headers
        for key, value in BLOCKED_HEADERS.items():
            if self.headers.get(key) == value:
                return True

        return False

def run(server_class=HTTPServer, handler_class=FirewallHandler,
port=8080):
    server_address = ('', port)
    httpd = server_class(server_address, handler_class)
    print(f"Firewall server running on port {port}...")
    httpd.serve_forever()

if __name__ == "__main__":
```

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```
run()
```